# The GAMA modeling and simulation platform

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**Abstract**

Agent-based modeling is now used in numerous application domains (ecology, social sciences, etc.) but its adoption is still impeded by the lack of generic yet ready-to-use tools supporting modelers in designing and simulating complex models integrating different modeling paradigms, multiple levels of agency and realistic environments. The open-source GAMA modeling and simulation platform, developed since 2007 by a consortium of public and private partners, has been designed to specifically address these three issues, while providing, at the same time, a complete model development environment accessible to novice modelers, an extensible library of reusable agent architectures and skills and a high-level visualization system.

**Keyword:** agent-based modeling, multi-scale modeling, simulation platform

## 1. Agent-based modeling platforms

In the last ten years there has been a striking increase in the use of agent-based models to study and simulate complex systems. This rise has been due, in part, to the availability of software platforms that have now reached a satisfying level of maturity, namely Netlogo [2] and Repast [3]. However, their success has also created a situation where the former, aimed at novice modelers, is limited, by its own design, to only allow the development of toy/prototype models, while the latter requires advanced programming skills to exploit all the features it provides. Scientists or industrials wishing to develop complex agent-based models (i.e. models with a rich set of input data, for instance GIS data, coupled with other legacy models, tackling multiple scales of representation, etc.) have then no choice but to rely on computer scientists to develop them.

## 2. GAMA

The GAMA (GIS & Agent-based Modeling Architecture) modeling platform [1] has been developed since 2007 by a consortium of public and private partners to address this problem. This open-source platform, adopted by about 2000 users as of 2014, provides a complete software environment for building multi-scale, spatially explicit, agent-based simulations. It supports different modeling paradigms in addition to agents (e.g., cellular automata, equation-based models), accepts a variety of input data (like GIS data, for example), provides an advanced visualization subsystem supporting multiple viewpoints on models and ships with ready-to-use abstractions for the most common needs (e.g., decision architectures, generic agent's skills such as movements regardless on the representation of the environment), which are accessible through a dedicated high-level modeling language (GAML) that remains extensible by Java programmers.

This talk will demonstrate the main features of the platform and showcase a number of projects that have been developed with it over the last seven years.

## References

[1] http://gama-platform.googlecode.com/

[2] https://ccl.northwestern.edu/netlogo/

[3] http://repast.sourceforge.net/